G12Adv	Mathematics	الرياضيات	T1 - 2020/21

Multiple Choice Part	جزء الاختيار من متعدد
Show all your work when answering these questions.	يجب كتابة خطوات الحل التفصيلية للمفردات الاختبارية كافة.

Multiple Choice — اختيار من متعدد

1. Find the derivative of the function $f(x) = \tan^{-1}(x^3)$.

A) $f'(x) = \dfrac{-1}{\sqrt{1-x^6}}$

B) $f'(x) = \dfrac{1}{\sqrt{1-x^6}}$

C) $f'(x) = \dfrac{3x^2}{1+x^6}$

D) $f'(x) = \dfrac{1}{1+x^6}$

2. Find the equation of the tangent line to the function $f(x) = \sqrt{x+3}$ at $x = -2$.

A) $y = 4(x+2)+2$

B) $y = \frac{1}{4}(x-1)+2$

C) $y = \frac{1}{2}(x+2)+1$

D) $y = \frac{1}{2}(x-2)+1$

3. If $h'(x) = n\sqrt{h(x)}$ where $n > 0$ and $h''(x) = 18$ at a given point x. Find the value of n.

A) 6 **B)** 36

C) 3 **D)** $3\sqrt{2}$

4. If $|g(x) - 4| \leq 2(2-x)$ is true for all values of x, evaluate $\lim_{x\to 2} g(x)$.

A) -4 **B)** 0

C) 2 **D)** 4

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5. **Given that** $\lim_{x\to\infty} \dfrac{3x^4 + mx^4 - 2x^3 - 1}{2x^4 + 2x^3 - x} = 4$, **find the constant** m.

A) -2

B) $\dfrac{1}{2}$

C) 4

D) 5

6. **Evaluate**

$$\lim_{x\to 0} \left(\frac{\frac{1}{x+2} - \frac{1}{2}}{x} \right)$$

A) 0

B) $-\dfrac{1}{4}$

C) $\dfrac{1}{4}$

D) Does not exist

7. **Evaluate**

$$\lim_{x\to 0} \left(\frac{1 - \cos x}{x \sin x} \right)$$

A) 0

B) $\dfrac{1}{2}$

C) 2

D) Does not exist

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Multiple Choice — اختيار من متعدد

8. Find all points for which the tangent line to $x^2 + y^2 - 2y = 0$ is horizontal.

A) $(0,0),\ (0,2)$ **B)** $(0,0),\ (0,-2)$

C) $(0,2)$ **D)** $(0,0)$

9. Find the derivative of the function $f(x) = e^x \ln x$.

A) $f'(x) = xe^x$

B) $f'(x) = \dfrac{e^x}{x} + e^x \ln x$

C) $f'(x) = \dfrac{e^x}{x} + \ln x$

D) $f'(x) = e^x + \dfrac{1}{x}$

10. A ball is thrown upward then moves according to the relation $s(t) = 56t - 4t^2$, where t is in seconds and s in meters. What is the maximum height the ball can reach?

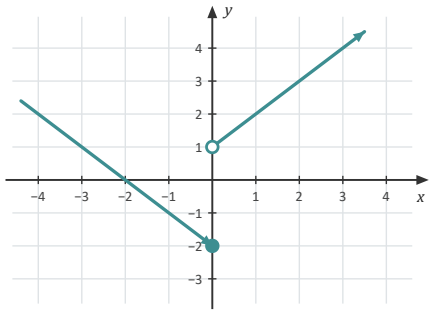
A) 7 m **B)** 168 m

C) 196 m **D)** 392 m

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Multiple Choice — اختيار من متعدد

11. Use the graph to determine $\lim_{x\to 0^+} f(x)$, if it exists.



A) -2 B) 1

C) 2 D) 3

12. If $f(x) = x^4 - 5x$ then $\lim_{x\to 2} \dfrac{f'(x) - f'(2)}{x - 2}$ equals

A) 6 B) 27

C) 38 D) 48

13. Determine the values of m and n that make the function $f(x)$ continuous at $x = 1$.

$$f(x) = \begin{cases} \dfrac{x^2 - mx + 2}{x - 1}, & x \neq 1 \\ n, & x = 1 \end{cases}$$

A) $m = -3,\ n = -1$ B) $m = -3,\ n = 1$

C) $m = 3,\ n = -1$ D) $m = 3,\ n = 1$

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14. **Find the derivative of the function** $f(x) = \dfrac{\sin x^2}{x^2}$.

A) $f'(x) = \dfrac{\cos x^2}{2x}$

B) $f'(x) = \dfrac{2x \cos x^2}{2x}$

C) $f'(x) = \dfrac{2x^3 \cos x^2 + 2x \sin x^2}{x^4}$

D) $f'(x) = \dfrac{2x^3 \cos x^2 - 2x \sin x^2}{x^4}$

15. **The function** $f(x) = \sqrt{|x-1|}$ **is defined for all** x**. Which of the following statements is true?**

A) $f(x)$ is continuous and not differentiable at $x = 1$

B) $f(x)$ is differentiable at $x = 1$

C) $f(x)$ is not continuous at $x = 1$

D) $\lim_{x \to 1} f(x) \neq 0$

16. **Find any slant asymptotes of** $f(x) = \dfrac{x^3}{x^2 + x - 4}$.

A) $y = x - 1$

B) $y = 4x - 5$

C) $y = 5x - 4$

D) $y = x + 1$

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17. **Suppose that the diameter of an animal's pupils is given by $f(x)$ mm, where x is the intensity of light on the pupils, $f(x) = \dfrac{160x^{-0.4} + 90}{4x^{-0.4} + 15}$. Find the diameter of the pupils with (a) minimum light (b) maximum light.**

A) Min = 6 mm , Max = 40 mm

B) Min = 40 mm , Max = 6 mm

C) Min = 3 mm , Max = 20 mm

D) Min = 20 mm , Max = 3 mm

18. **For $f(x) = \begin{cases} x^2 + 2x, & x < 0 \\ ax + b, & x \geq 0 \end{cases}$ find all real numbers a and b such that $f'(0)$ exists.**

A) $a = -2,\ b = 0$

B) $a = 0,\ b = -2$

C) $a = 0,\ b = 2$

D) $a = 2,\ b = 0$

19. **Use the given position function $s(t) = \sqrt{t} + 2t^2$ to find the acceleration at $t = 1$.**

A) $-\dfrac{15}{2}$

B) $\dfrac{15}{2}$

C) $\dfrac{15}{4}$

D) $-\dfrac{15}{4}$

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20. Determine the value(s) of x for which the tangent line to $y = x^3 - 3x + 1$ is at an angle of 45° with the x-axis, assuming that angles are measured counterclockwise.

A) $x = -\frac{2}{\sqrt{3}}$

B) $x = \frac{2}{\sqrt{3}}$

C) $x = \pm\frac{2}{\sqrt{3}}$

D) $x = \pm\frac{4}{3}$